Detecting AI vs Real Images using Machine Learning

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Introduction

The ability to distinguish between AI-generated and real images has become increasingly relevant with the rapid improvement of generative models. While modern generative systems can produce highly realistic images, subtle statistical and structural differences still exist, making this a natural problem for machine learning.

The goal of this project is to compare several machine learning approaches for binary image classification, specifically identifying whether an image is real or AI-generated. In addition to comparing model performance, this study aims to understand how model structure, particularly the ability to capture spatial information, affects classification accuracy.

The dataset used is the *AI vs Real Images Dataset* obtained from Kaggle. While the dataset description suggests approximately 1500 images, preprocessing revealed 995 usable images, with a noticeable imbalance between real and AI-generated samples. The data spans multiple categories including animals, city scenes, food, nature, and people.

Methods

Dataset and Preprocessing

All images were resized to 64×64 to reduce computational cost. Each image was normalized to the range $[0, 1]$. The dataset was split into training and testing sets using an 80/20 split.

Images were flattened into vectors for classical models, while convolutional neural networks operated directly on the image tensors.

Dimensionality Reduction (PCA)

Due to the high dimensionality of image data, Principal Component Analysis (PCA) was applied prior to training the classical models. Given a data matrix $X \in R^{n \times d}$, PCA finds a lower-dimensional representation by solving:

$$\max_{\mathrm{v}_1,\ldots,\mathrm{v}_k} \sum_{i=1}^{k} \mathbf{v}_i^T S \mathbf{v}_i$$

subject to $\|\mathbf{v}_i\| = 1$ and $\mathbf{v}_i^T \mathbf{v}_j = 0$ for $i \neq j$, where S is the sample covariance matrix. The data is then projected as:

$$Z = XV_k$$

where V_k contains the top k eigenvectors.

Models

1. Logistic Regression

Logistic regression models the probability of a class using:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. The parameters are learned by minimizing the cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{n} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

2. Support Vector Machine (SVM)

The SVM seeks a separating hyperplane by solving:

$$\min_{\mathrm{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{n} \xi_i$$

subject to:

$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

3. Convolutional Neural Networks (CNN)

CNNs operate directly on image data by applying convolutional filters. A convolutional layer computes:

$$h_{i,j}^{(k)} = \sigma\left(\sum_{m,n} w_{m,n}^{(k)} x_{i+m,j+n} + b^{(k)}\right)$$

where $w^{(k)}$ is the k-th filter. Pooling layers reduce spatial dimensions:

$$h'_{i,j} = \max_{(m,n)\in\mathcal{R}} h_{i+m,j+n}$$

The network is trained by minimizing cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{n} y_i \log(\hat{y}_i)$$

using gradient-based optimization (Adam).

Two CNN architectures were implemented:

- A baseline CNN with two convolutional layers
- An improved CNN with an additional convolutional layer and dropout regularization

Implementation

All models were implemented in Python using a Jupyter notebook environment. Image preprocessing was performed using standard libraries such as NumPy and PIL, while classical models were implemented using `scikit-learn` and convolutional neural networks were implemented using `PyTorch`.

Preprocessing

The preprocessing steps described in the Methods section were applied prior to model training. These processed images were then used as inputs for both classical and neural network models, with classical models operating on flattened feature vectors.

Dimensionality Reduction

Due to the high dimensionality of the flattened image vectors, Principal Component Analysis (PCA) was applied prior to training logistic regression and SVM models. PCA was implemented using `scikit-learn`, reducing the input dimension to 100 principal components. This corresponds to projecting the original data onto the top eigenvectors of the covariance matrix, as described in the Methods section.

Logistic Regression and SVM

Logistic regression and SVM models were trained using the PCA-transformed data. Logistic regression was optimized using a cross-entropy loss function with iterative solvers provided by `scikit-learn`. The SVM was implemented using a linear kernel, solving the margin optimization problem described earlier.

Both models operate on the reduced feature representation and treat each input as a vector, without explicitly accounting for spatial structure in the original image.

Convolutional Neural Networks

Unlike the classical models, CNNs were implemented directly on the image tensors using `PyTorch`. Each input image was represented as a $64 \times 64 \times 3$ tensor, preserving spatial structure.

The baseline CNN consisted of two convolutional layers followed by max-pooling operations, a fully connected layer, and a final output layer. The improved CNN extended this architecture by adding an additional convolutional layer and incorporating dropout regularization in the fully connected stage.

The networks were trained using the Adam optimizer with a cross-entropy loss function. Training was performed over multiple epochs, and model performance was evaluated on a held-out test set. The use of dropout in the improved model helps reduce overfitting by randomly deactivating neurons during training.

Evaluation

Model performance was evaluated using classification accuracy, confusion matrices, and classification reports. These metrics provide insight into both overall performance and class-specific behavior, particularly in the presence of class imbalance.

Results

Test Accuracies

The following accuracies were obtained on the test set:

- Logistic Regression: 75.4%
- SVM: 78.9%
- CNN (baseline): 85.9%
- CNN (improved): 87.9%

The CNN-based models clearly outperform the classical approaches. The improved CNN shows a modest but consistent improvement over the baseline model.

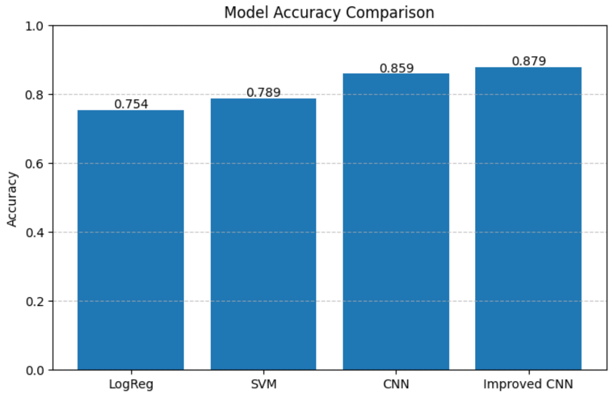

Figure 1: Comparison of classification accuracy across models.

Confusion Matrices

The classification reports for logistic regression and SVM indicate strong performance on real images but poor recall for AI-generated images, with recall values near 0.26. The CNN models significantly improve detection of AI-generated images, though performance remains lower than for real images.

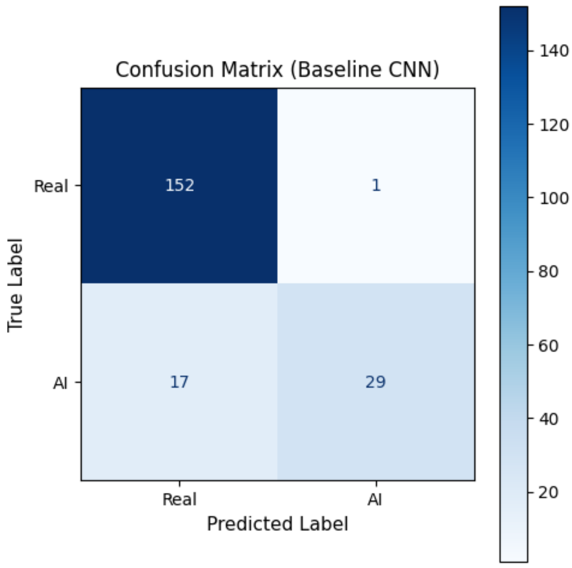

Figure 2: Baseline CNN confusion matrix.

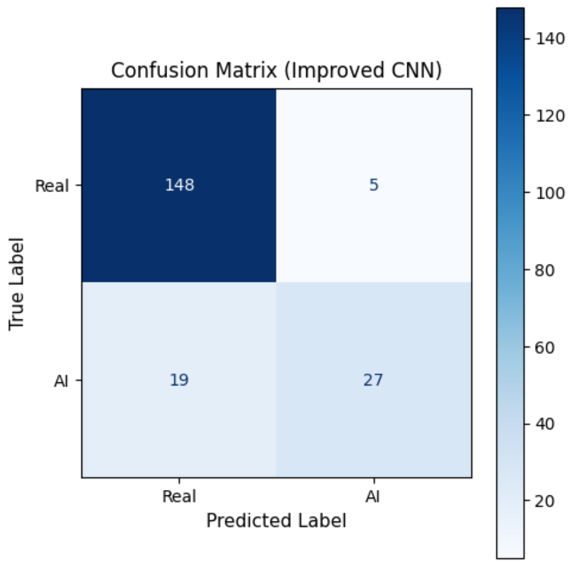

Figure 3: Improved CNN confusion matrix.

Qualitative Analysis of Model Predictions

To better understand model behavior beyond aggregate accuracy, several representative examples from the test set were examined. These examples were grouped based on how different models performed on them.

Hard Failures (All Models Incorrect)

These examples highlight a consistent limitation across all models. In both cases, the images exhibit highly realistic lighting, smooth gradients, and natural structure (e.g., landscapes and reflections). Such features make the images visually indistinguishable from real photographs, suggesting that current models struggle when AI-generated content closely mimics real-world distributions.

Figure 4: Examples where all models misclassified AI-generated images as real.

CNN Advantage (Only CNN Models Correct)

In these cases, convolutional neural networks successfully identified subtle inconsistencies that were missed by logistic regression and SVM. These images often contain complex textures, such as fur or fine-grained object boundaries. This supports the idea that CNNs benefit from their ability to capture local spatial relationships, which are lost when images are flattened into vectors for classical models.

Figure 5: Examples where CNN-based models correctly classified images while classical models failed.

Easy Cases (All Models Correct)

These images contain clear and distinguishable visual features, making them easier to classify. In such cases, even simple models are able to perform well, indicating that strong and unambiguous patterns reduce the need for more complex feature extraction.

Figure 6: Examples where all models correctly classified the images.

CNN Failure Cases

Interestingly, there are also cases where CNNs perform worse than simpler models. These images tend to have structured layouts or repetitive patterns, such as arranged objects or faces. In such scenarios, CNNs may overfit to local patterns and fail to capture broader structural cues, while classical models relying on global features may perform better.

Figure 7: Examples where CNN models misclassified images while classical models performed better.

Overall, this qualitative analysis reinforces the quantitative results and demonstrates that model performance depends strongly on the visual characteristics of the input data.

Discussion

Model Performance

The results of this study show a clear advantage for convolutional neural networks in the task of distinguishing between real and AI-generated images. Both CNN models significantly outperformed logistic regression and SVM, achieving accuracies of approximately 86% and 88%, respectively. This improvement is consistent with the underlying structure of the models. CNNs explicitly exploit spatial relationships through convolutional filters and pooling operations, whereas logistic regression and SVM operate on flattened representations of the image, effectively discarding spatial information. These findings are also consistent with prior work in computer vision, where convolutional neural networks have been shown to outperform classical machine learning approaches on image classification tasks (*LeCun et al., 1998*).

Effect of Dimensionality Reduction

Although PCA was used to reduce dimensionality for the classical models, this transformation compresses the data and may remove subtle features that are important for classification. This likely contributes to the lower performance observed for logistic regression and SVM. In contrast, CNNs are able to learn hierarchical representations directly from the image data, making them better suited for this type of problem.

Model Complexity

The comparison between the baseline and improved CNN further highlights the role of model complexity. The improved CNN, which incorporates an additional convolutional layer and dropout, achieved a modest but consistent increase in performance. This suggests that increasing model capacity can improve results, although the gains appear limited by the size and quality of the dataset. This observation is consistent with more recent developments in deep learning, where increasing network depth and model capacity has been shown to improve performance on complex image classification tasks (*Krizhevsky et al., 2012*).

Qualitative Analysis

Beyond overall accuracy, the qualitative analysis of individual predictions provides additional insight into model behavior. Several patterns emerged from examining representative examples. In some cases, all models failed to correctly classify highly realistic AI-generated images, particularly those with smooth lighting, natural textures, and realistic structure. These examples highlight the inherent difficulty of the problem when synthetic images closely resemble real-world data.

In other cases, CNN-based models were able to correctly classify images that classical models misclassified. These images often contained complex textures, such as fur or fine object boundaries, suggesting that CNNs benefit from their ability to capture local spatial features. However, CNNs were not universally superior. There were also instances where CNN models misclassified images with structured or repetitive patterns, while simpler models produced correct predictions. This indicates that increased model complexity does not always guarantee improved performance across all types of images.

Class Imbalance

Finally, class imbalance plays a significant role in model performance. The dataset contains substantially more real images than AI-generated ones, leading to a bias toward predicting the majority class. This is reflected in the classification metrics, where all models show reduced recall for AI-generated images. Overall, these results suggest that model performance depends not only on architecture, but also on the visual characteristics of the data and the distribution of classes.

Limitations

Dataset Size and Imbalance

There are several limitations associated with this study that impact both model performance and the interpretation of results. First, the dataset is relatively small and exhibits a noticeable class imbalance, with significantly more real images than AI-generated images. This imbalance affects the learning process, as models are biased toward the majority class, leading to reduced recall for AI-generated images.

Image Resolution

Second, the preprocessing step of resizing all images to 64×64 introduces a loss of fine-grained detail. While this reduction was necessary for computational efficiency, it may remove subtle visual cues that are important for distinguishing between real and synthetic images. Higher-resolution inputs could potentially improve performance, particularly for models that rely on texture-based features.

Model Simplicity

Another limitation is the simplicity of the models used, particularly the convolutional neural networks. While the improved CNN demonstrated better performance than the baseline model, both architectures remain relatively shallow compared to modern deep learning approaches. More advanced architectures, such as deeper convolutional networks or transfer learning with pretrained models, could yield improved results.

Handling Class Imbalance

Additionally, no explicit techniques were used to address class imbalance. Methods such as resampling, class weighting, or data augmentation could potentially improve the detection of AI-generated images by providing more balanced training data.

Dataset Complexity

Finally, the dataset itself introduces limitations. The images span multiple categories, and the quality and realism of AI-generated images vary significantly. As observed in the qualitative analysis, some AI-generated images are nearly indistinguishable from real images, making the classification task inherently difficult. This suggests that further improvements may require not only better models, but also more diverse and higher-quality datasets.

References

- *AI vs Real Images Dataset*
- scikit-learn documentation
- PyTorch documentation
- Y. LeCun et al., "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, 1998.
- A. Krizhevsky et al., "ImageNet classification with deep convolutional neural networks," *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.